

Uncertainty-guided decoding: what the three branches actually contain

A lineage-aware review of **new-exp2**, **codex-recom-iter1** (the repository's actual name for 'codex-iter1'), and **MedMCQA**. The immediate question is deliberately narrow: does correctly oriented Inf-Log-Var predict whether an answer is wrong?

Bottom line

The branches contain increasingly strong experimental machinery, but none contains a committed split manifest, trained weights, or MedExQA/MedMCQA result files. Therefore no branch currently demonstrates that ILV predicts correctness. Use **MedMCQA as the code base**, repair four small trust gaps, and run a minimal fixed-sign pilot before any large ablation suite.

Branch	Role	Current verdict
new-exp2	First MedExQA generation prototype	Historical prototype only
codex-recom-iter1	Scientific protocol repair	Best methods scaffold
MedMCQA	Scale-up plus safer orchestration	Recommended implementation base

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Scope: committed branch contents plus ignored local project notes; read-only inspection; no training run was executed.

1. Executive summary

Decision

Continue from **MedMCQA**, not from the other two branches. It is a strict descendant of codex-recom-iter1 and already includes the online ILV, fixed-sign statistics, larger dataset, step-based validation, provenance counts, and a wrapper that pauses before test evaluation.

- Do **not** interpret the branch history as three independent experiments. Each later branch inherits the earlier one.
- The minimal thesis claim is still untested: **higher ILV -> higher probability of a wrong answer**. An AUROC below 0.5 is a negative result, not a score to flip.
- The cleanest immediate endpoint is a generated, machine-readable answer followed by a free-text explanation. This avoids regex-only grading and avoids silently dropping uncertain non-answers.
- BioASQ remains the stronger confirmatory dataset because exact answers can label knowledge while ideal answers preserve natural-language generation. MedMCQA is suitable for the faster interim MCQ-plus-explanation result.

Branch lineage

new-exp2	codex-recom-iter1	MedMCQA
MedExQA generation prototype 9 commits from 94700a9	Protocol and analysis repairs 21 commits beyond new-exp2	MedMCQA scale-up 5 commits beyond codex-recom-iter1

new-exp2 -> codex-recom-iter1 -> MedMCQA

What was actually verified

Check	Finding
Python syntax	All tracked Python files compile on all three branch snapshots.
Shell syntax	All tracked .sh files pass bash -n on all three branch snapshots.
Scientific outputs	No tracked MedExQA/MedMCQA results, weights, logs, or split manifests.
Local ignored files	41 results-debug files exist, but they are OBQA debug artifacts and are not branch-scoped evidence for this question.

2. At-a-glance comparison

Question	new-exp2	codex-recom-iter1	MedMCQA
Training data	MedExQA, about 740 derived train rows	Same MedExQA split, now with manifest machinery	MedMCQA gold explanations: 30k train / 1k val / 1k test
Generated answer label	Separate letter probe	Regex option commitment, then probe fallback	Regex option commitment only by default; non-commitments excluded
ILV timing	Post-hoc second pass	Online during generation is primary	Online during generation is primary
Routing readout	Deterministic default	Stochastic S=35 primary; deterministic ablation	Stochastic S=35 primary; deterministic ablation
Direction	May flip inverted AUROC	Fixed: higher ILV predicts wrong	Fixed: higher ILV predicts wrong
Statistics	Point estimates and fragile token permutation	Bootstrap CIs, circular-shift null, frozen val thresholds	Same, plus per-subject summaries
Reproducibility	Weak	Strong design, but manifest absent	Better wrapper, but manifest absent
End-to-end run	Loose launchers	Large driver; default can continue into test too early	Stateful wrapper pauses for manual audit
Evidence today	None tracked	None tracked	None tracked

Interpretation

The move from new-exp2 to codex-recom-iter1 is the major scientific improvement. The move from codex-recom-iter1 to MedMCQA is mainly a dataset and orchestration improvement. It does not yet solve the deeper label question: MedMCQA officially scores option choice, not free-form medical correctness.

3. new-exp2

PROTOTYPE

First serious attempt to move FCVR/ILV from multiple-choice training to explanation generation.

What it contains

- MedExQA data loading and Stage-1 generation fine-tuning with prompt-masked explanation loss.
- FCVR training, early stopping, best-checkpoint saving, and MedExQA launchers.
- Teacher-forced and greedy-generation token analysis: ILV, LM entropy, surprisal, token categories, correlations, spike plots, and HTML output.
- A sequence-level AUROC that compares ILV aggregates against entropy and NLL, using a separate A/B/C/D probe as the correctness label.
- An input-level ID-versus-ODD bridge test and same-domain OOD filtering.

What is missing or unreliable

Issue	Why it matters for the thesis
Wrong signal orientation policy	It reports 1 - AUROC when ILV points the other way. This chooses the direction after seeing evaluation data and can turn noise into apparent success.
Not decoding-time ILV	Generated text is read again in a second teacher-forced pass. The score is not the one available during the token decision.
Deterministic routing is primary	Changing earlier expert choices can change later hidden states and later ILV. S=35 stochastic routing should be primary.
Wrong primary endpoint	The model's generated explanation may be wrong while the separate letter probe is right, or vice versa.
No protected validation/test protocol	The branch analyzes test directly; thresholds, direction, and interpretations are not frozen on validation.
Fragile null and verdict	Independent token shuffling destroys serial structure; arbitrary effect thresholds remain in the verdict code.
No branch-scoped results	The implementation exists, but no committed output demonstrates a relationship.

Use it for

Historical understanding and small utilities only. Do not use its AUROC verdicts as thesis evidence, and do not restart new experiments from this branch.

4. codex-recom-iter1

METHODS BASE

The main protocol-repair branch. This is the repository's actual name for the branch described as 'codex-iter1'.

What it adds

Area	Added capability
Provenance	Applied seeds, a derived-split manifest format, run tags, collision checks, and val/test separation.
Correctness	Generated-text option commitment, independent probe/NLI/truncation fields, quality metrics, and a human-audit CSV.
Signal capture	Online ILV hooks during generation; post-hoc ILV retained only as a comparison.
Routing	Paper-style S=35 stochastic routing as primary; deterministic routing as an explicit ablation.
Baselines	Deterministic router, untrained FCVR, beta=0, entropy, NLL, length, gate entropy, prior and KL-mask arms.
Statistics	Fixed-sign AUROC, bootstrap CIs, circular-shift null, val-selected thresholds, test-only evaluation, AURC and residual models.
Future policy	Gated calibration scaffold; guided decoding itself is intentionally not implemented.

Remaining gaps

- The default primary label mixes two outcomes: generated-option correctness and a separate probe fallback. MedMCQA later fixes this default.
- The frozen configuration hash omits source/dataset, adapter identity, label policy, and some other run identity fields. Incompatible val and test files can therefore pass the guard.
- The default driver includes val and test in one run. It can enter test before the audit CSV is inspected or imported, despite the written protocol.
- The OOD length-matched control uses inclusive bounds on both sides of every bin; boundary examples can be sampled in two bins.
- MedExQA is very small for this training design: about 740 training, 50 validation, and 175 test examples after pooling the official splits.
- There are no tracked manifests, weights, or results. Training-seed sweeps and guided decoding are explicitly not done.

Assessment

This branch contains the best scientific logic of the three. Its main contribution is not a result; it is the protocol that makes a result believable.

5. MedMCQA

[USE THIS BASE](#)

A strict descendant of codex-recom-iter1 that scales the generation training and makes the run order safer.

What it adds

- A MedMCQA generation view using the question, four options, and the gold explanation as the free-text target.
- Filtered and stratified 30,000 train / 1,000 validation / 1,000 test design, with deduplication, subject metadata, explanation-length limits, and overlap checks.
- Step-based validation and early stopping for the much larger dataset, plus max-sequence-length filtering.
- MedMCQA launchers, a full driver, and a stateful sequence wrapper that stops after validation for the required human audit.
- A safer default label: only generated-text option commitment is primary; probe fallback is opt-in and label provenance is counted.
- Per-subject reporting and a guard against using the overlapping MedMCQA-derived dataset as OOD.

What it still does not establish

Limitation	Plain-language consequence
MedMCQA is still an MCQ task	The prompt includes options. The experiment measures explanation generation around option selection, not fully option-free medical answering.
The explanation is not a correctness key	The exp field is a single, potentially noisy rationale. Similarity to it cannot prove that arbitrary generated medical prose is correct.
Non-commitments are dropped	If uncertain generations omit an answer letter, the primary AUROC may be calculated on an easier, selected subset.
First-token window is not answer-aware	The code can summarize early ILV, but it does not detect the exact commitment position. A first-10 window may already include the answer.
Split is not frozen in git	Manifest tooling exists, but no splits/medmcqa_gen-derived-seed42.csv is committed.
Hyperparameters are inherited	Learning rate, batch size, and warmup were carried over from MedExQA rather than tuned for a roughly 40x larger corpus.
No evidence artifacts	There is still no trained adapter, FCVR checkpoint, labeled readout, or final AUROC in the branch.

Recommended interpretation

Use MedMCQA for an **interim** claim: ILV predicts whether the model selected the correct option while generating an explanation. Do not call this validated free-form medical correctness.

6. The eight major pitfalls and where they stand

#	Pitfall	new-exp2	codex-recom-iter1	MedMCQA / action
1	Wrong correctness label	Separate probe labels another pass	Generated option added, but probe fallback mixed in	Option-only default. Better: force an explicit answer field.
2	Post-hoc rather than online ILV	Primary score is a second pass	Online hooks added	Resolved for primary readout
3	Deterministic routing changes the trajectory	Deterministic default	S=35 stochastic primary	Resolved; keep deterministic as ablation
4	No fair baselines	Entropy/NLL only	Full ladder plus gate entropy and length	Implemented; postpone full ladder until signal pilot works
5	OOD or prompt-format confounds	Partial same-domain filtering	Canonical templates, format and length controls	Mostly implemented; fix duplicate bin boundaries
6	Inverted AUROC treated as success	Flips direction	Fixed sign in abstention report	Resolved for primary analysis; remove old inverted verdict language
7	Leakage and irreproducible splits/runs	Weak	Manifest and collision machinery	Wrapper improved, but the actual manifest is still absent
8	Exploratory statistics presented as confirmation	Arbitrary verdicts and token shuffle	CLs, circular shift, val freeze	Resolved in design; no confirmatory run exists yet

The one unresolved conceptual pitfall

A model that does not state an answer may be exactly the model that is most uncertain. Dropping those rows can make ILV look better or worse. The clean solution is not a more complex regex; it is a required structured answer prefix.

7. Fix these before spending GPU time

Priority	Fix	Minimum acceptance check
P0	Commit the MedMCQA split manifest and dataset revision/digest.	Fresh checkout reproduces the same 30k/1k/1k hashes; missing manifest must fail, not warn.
P0	Require generated output: ANSWER: then EXPLANATION: .	Parser coverage near 100%; invalid/missing answers are explicitly wrong or abstained by a predeclared rule, never silently dropped.
P0	Expand the frozen-config hash.	Include source, dataset, adapter/checkpoint digest, seed policy, label policy, prompt template, split-manifest digest, routing and generation settings.
P0	Make test entry require an audit-complete marker.	The runner cannot start test immediately after val unless the human audit is imported and frozen.
P1	Fix inclusive OOD length bins.	Use half-open bins except the final bin; assert no repeated indices.
P1	Assert answer tokenization for the secondary letter probe.	Tokenize the exact continuation in context; fail if A-D do not map as assumed.
P1	Delete or quarantine inverted-verdict language.	Every primary report states: higher ILV predicts wrong; AUROC below 0.5 is a negative result.
P2	Stop maintaining duplicate training scripts.	One canonical implementation is called by launchers; the root/scripts copies cannot drift.

Scope control

- Do not start MAP-prior, KL-mask, OOD, calibration, or expert-LoRA sweeps until the main ILV-versus-wrong pilot produces a stable, correctly oriented effect.
- Do not use test to choose ILV aggregation, sign, answer policy, beta, prompt, or thresholds. Choose them on validation; touch test after the audit and freeze.
- Do not claim the model 'knows medicine' from explanation similarity. The immediate defensible label is whether its explicit selected answer matches the gold option.

8. Recommended next experiment

Primary question

On held-out MedMCQA questions, does **online ILV averaged over the final 10 generated tokens** rank explicitly wrong answers above correct answers, with the direction fixed in advance?

Minimal pilot - do this first

Step	Action	Output / stop rule
1	Apply the four P0 fixes above on MedMCQA.	A small unit-tested data/label/provenance patch.
2	Run preflight and commit the manifest. Inspect dataset funnel, target lengths, masks, and 30-50 formatted examples.	Stop if answers are not reliably parseable or split hashes are not frozen.
3	Train one Stage-1 adapter and one FCVR arm: pretrained prior, beta=0.01, attention-masked KL, training seed 42.	Keep best validation NLL checkpoint. Record exact weight digest.
4	Run validation only: S=35 stochastic routing, inference seed 42, 200-500 examples, max 256 tokens.	Audit 50 outputs. Freeze label policy, score, threshold logic, and configuration.
5	Primary analysis on validation: <code>ilv_online_mean_last10</code> -> wrong. Report AUROC and 95% bootstrap CI without flipping.	Proceed only if pipeline is correct and both outcome classes have enough examples; a null result is acceptable.
6	Add minimum baselines: entropy max, NLL/token, output length, and gate entropy. Fit a val-only baseline model, then add ILV.	ILV should add ranking or likelihood information beyond cheap baselines.
7	After audit/freeze, run the untouched 1,000-example test once for the selected configuration.	Primary result: fixed-sign test AUROC [CI], accuracy, judge/parse coverage, group score summaries, and AURC.
8	Repeat inference seeds 43 and 44 as robustness checks, not new selection opportunities.	Report mean, range, and per-seed results.

Required primary report

- N total, N correct, N wrong, N invalid/uncommitted, and exactly how invalid outputs were treated.
- ILV distributions by correctness, fixed-sign AUROC with 95% example-level bootstrap CI, and AURC/risk-coverage.
- The same statistics for entropy, NLL, length, and gate entropy, plus a val-fitted baseline-plus-ILV comparison.
- One table of failure cases: high-ILV correct, low-ILV wrong, parse failures, truncations, and contradictions.

9. Path from this pilot to uncertainty-guided decoding

Gate	Question	If yes	If no
A. Association	Does correctly oriented online ILV predict wrong answers?	Freeze signal and aggregation.	Diagnose layer/token aggregation, training, and label noise. Do not decode with it.
B. Incremental value	Does ILV add value beyond entropy, NLL, length, and gate entropy?	Fit a val-only error probability model.	Prefer the simpler baseline; ILV is not justified yet.
C. Calibration	Is $P(\text{wrong} \mid \text{ILV, baselines})$ calibrated on held-out data?	Define policy thresholds and costs.	Recalibrate or restrict the operating region.
D. Intervention	Can a causal decoding intervention improve quality at fixed cost/coverage?	Compare guided decoding, abstention, and standard decoding on frozen test sets.	Treat ILV as a diagnostic only.
E. Generalization	Does the relationship transfer to a cleaner natural-language task?	Move to BioASQ exact+ideal answers, then K-QA external evaluation.	Report task dependence; avoid universal uncertainty claims.

Recommended dataset sequence

Now: MedMCQA with explicit answer plus explanation, because the code is almost ready and the gold option gives a clean interim label. **Confirmatory:** BioASQ questions with both `exact_answer` and `ideal_answer`, because exact-answer accuracy/F1 measures knowledge while the ideal answer remains a genuine NLG target.

External/OOD: K-QA as a small, physician-authored long-form test.

Repository evidence reviewed

Item	Evidence
Branch tips	new-exp2 5ccc5c0; codex-recom-iter1 4b9a7a0; MedMCQA cc7ac01
Lineage deltas	94700a9...new-exp2: 16 files; new-exp2...codex-recom-iter1: 38 files; codex-recom-iter1...MedMCQA: 20 files
Tracked protocol note	codex-recom-iter1-notes.md
Ignored project notes	medmcqa-notes.md; medmcqa-primary-source-notes.md; medical-nlg-dataset-recommendations.md
Core code	analyze_token_signals.py; label_generation_correctness.py; abstention_report.py; baseline_ladder_report.py; utils/data.py; run-iter1-granite-*.sh; run-medmcqa-sequence.sh
Review limits	Static and design review only. No GPU training, dataset download, model inference, or numeric claim verification.

Note: the repository does not contain docs/agents/issue-tracker.md, so this review used the branch notes and the stated thesis goal as the experiment specification.